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² Plasticity, nutrition, and breeding
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First, we model external conditions as a normal distribution

$$C(y) = \frac{1}{\sqrt{2\pi w^2}} \exp\left(-\frac{y^2}{2w^2}\right)$$

And fecundity as a linear function:

$$F(n, y) = a_f n + cy.$$

Then, the fitness is given by:

$$W(n, y) = C(y)F(n, y) = \frac{1}{\sqrt{2\pi w^2}} \exp\left(-\frac{y^2}{2w^2}\right) (a_f n + cy)$$

To find the optimal breeding date, we take the partial derivative against y :

$$\frac{\partial W(n, y)}{\partial y} = \frac{1}{\sqrt{2\pi w^2}} \exp\left(-\frac{y^2}{2w^2}\right) \times (c - y(a_f n + cy)/w^2)$$

Setting this equal to zero, we have:

$$y = \frac{-a_f n + \sqrt{(a_f n)^2 + 4c^2 w^2}}{2c}$$

Now, if we average the fitness $W(n, y)$ over the breeding date distribution y , we

get the mean fitness:

$$\begin{aligned}
\bar{W}(n, \bar{y}) &= \int p(y)W(n, y)dy \\
&= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma_y^2}} \exp\left(-\frac{(y-\mu)^2}{2\sigma_y^2}\right) \times \frac{1}{\sqrt{2\pi w^2}} \exp\left(-\frac{y^2}{2w^2}\right) (a_f n + cy) dy \\
&= \frac{1}{2\pi\sigma_y\omega} \int_{-\infty}^{\infty} \exp\left(-\frac{(y-\mu)^2}{2\sigma_y^2}\right) \times \exp\left(-\frac{y^2}{2w^2}\right) (a_f n + cy) dy \\
&= \frac{1}{2\pi\sigma_y\omega} \int_{-\infty}^{\infty} \exp\left(-\frac{w^2(y-\mu)^2 + \sigma_y^2 y^2}{2\sigma_y^2 w^2}\right) (a_f n + cy) dy \\
&= \frac{1}{2\pi\sigma_y\omega} \int_{-\infty}^{\infty} \exp\left(-\frac{w^2(y^2 - 2\mu y + \mu^2) + \sigma_y^2 y^2}{2\sigma_y^2 w^2}\right) (a_f n + cy) dy \\
&= \frac{1}{2\pi\sigma_y\omega} \int_{-\infty}^{\infty} \exp\left(-\frac{(w^2 + \sigma_y^2)y^2 - 2w^2\mu y + w^2\mu^2}{2\sigma_y^2 w^2}\right) (a_f n + cy) dy \\
&= \frac{1}{2\pi\sigma_y\omega} \exp\left(-\frac{w^2\mu^2}{2\sigma_y^2 w^2}\right) \int_{-\infty}^{\infty} \exp\left(-\frac{(w^2 + \sigma_y^2)y^2 - 2w^2\mu y}{2\sigma_y^2 w^2}\right) (a_f n + cy) dy \\
&= \frac{1}{2\pi\sigma_y\omega} \exp\left(-\frac{w^2\mu^2}{2\sigma_y^2 w^2}\right) \\
&\quad \times \int_{-\infty}^{\infty} \exp\left(-\frac{y^2 - 2w^2\mu y/(w^2 + \sigma_y^2) + (w^2\mu/(w^2 + \sigma_y^2))^2 - (w^2\mu/(w^2 + \sigma_y^2))^2}{2\sigma_y^2 w^2/(w^2 + \sigma_y^2)}\right) \\
&\quad \times (a_f n + cy) dy \\
&= \frac{1}{2\pi\sigma_y\omega} \exp\left(-\frac{w^2\mu^2}{2\sigma_y^2 w^2}\right) \\
&\quad \times \int_{-\infty}^{\infty} \exp\left(-\frac{(y - w^2\mu/(w^2 + \sigma_y^2))^2 - (w^2\mu/(w^2 + \sigma_y^2))^2}{2\sigma_y^2 w^2/(w^2 + \sigma_y^2)}\right) \\
&\quad \times (a_f n + cy) dy \\
&= \frac{1}{2\pi\sigma_y\omega} \exp\left(-\frac{w^2\mu^2}{2\sigma_y^2 w^2}\right) \\
&\quad \times \int_{-\infty}^{\infty} \exp\left(-\frac{(y - w^2\mu/(w^2 + \sigma_y^2))^2}{2\sigma_y^2 w^2/(w^2 + \sigma_y^2)}\right) \exp\left(\frac{w^2\mu^2}{2\sigma_y^2(w^2 + \sigma_y^2)}\right) (a_f n + cy) dy \\
&= \frac{1}{2\pi\sigma_y\omega} \exp\left(-\frac{\mu^2}{2\sigma_y^2}\right) \exp\left(\frac{w^2\mu^2}{2\sigma_y^2(w^2 + \sigma_y^2)}\right) \\
&\quad \times \int_{-\infty}^{\infty} \exp\left(-\frac{(y - w^2\mu/(w^2 + \sigma_y^2))^2}{2\sigma_y^2 w^2/(w^2 + \sigma_y^2)}\right) (a_f n + cy) dy \\
&= \frac{1}{2\pi\sigma_y\omega} \exp\left(-\frac{\mu^2}{2(w^2 + \sigma_y^2)}\right) \\
&\quad \times \int_{-\infty}^{\infty} \exp\left(-\frac{(y - w^2\mu/(w^2 + \sigma_y^2))^2}{2\sigma_y^2 w^2/(w^2 + \sigma_y^2)}\right) (a_f n + cy) dy
\end{aligned}$$

We still need to spend more time doing the integrals:

$$\begin{aligned}
\bar{W}(n, \bar{y}) &= \frac{1}{2\pi\sigma_y\omega} \exp\left(-\frac{\mu^2}{2(w^2 + \sigma_y^2)}\right) \\
&\times \frac{\sqrt{2\pi\sigma_y^2 w^2/(w^2 + \sigma_y^2)}}{\sqrt{2\pi\sigma_y^2 w^2/(w^2 + \sigma_y^2)}} \int_{-\infty}^{\infty} \exp\left(-\frac{(y - w^2\mu/(w^2 + \sigma_y^2))^2}{2\sigma_y^2 w^2/(w^2 + \sigma_y^2)}\right) \\
&\times (a_f n + cy) dy \\
&= \frac{1}{\sqrt{2\pi(w^2 + \sigma_y^2)}} \exp\left(-\frac{\mu^2}{2(w^2 + \sigma_y^2)}\right) \\
&\times \frac{1}{\sqrt{2\pi\sigma_y^2 w^2/(w^2 + \sigma_y^2)}} \int_{-\infty}^{\infty} \exp\left(-\frac{(y - w^2\mu/(w^2 + \sigma_y^2))^2}{2\sigma_y^2 w^2/(w^2 + \sigma_y^2)}\right) \\
&\times (a_f n + cy) dy \\
&= \frac{1}{\sqrt{2\pi(w^2 + \sigma_y^2)}} \exp\left(-\frac{\mu^2}{2(w^2 + \sigma_y^2)}\right) \left(a_f n + \frac{cw^2\mu}{w^2 + \sigma_y^2}\right)
\end{aligned}$$

If we substitute $\gamma = 1/(w^2 + \sigma_y^2)$ and $\mu = \bar{y}$, then,

$$\bar{W}(n, \bar{y}) = \sqrt{\frac{\gamma}{2\pi}} \exp\left(-\frac{\gamma \bar{y}^2}{2}\right) (a_f n + \gamma c w^2 \bar{y})$$

If we write $\bar{y} = \bar{a} + \bar{b}n$, then

$$\begin{aligned}\beta &= \left(\frac{\partial/\partial \bar{a}}{\partial/\partial \bar{b}} \right) \ln \bar{W}(n, \bar{y}) \\ &= \left(\frac{\partial/\partial \bar{a}}{\partial/\partial \bar{b}} \right) \left(\ln \frac{\gamma}{2\pi} - \frac{\gamma(\bar{a} + \bar{b}n)}{2} + \ln(a_f n + \gamma c w^2(\bar{a} + \bar{b}n)) \right) \\ &= \left(\frac{-\frac{\gamma}{2} + \frac{\gamma c w^2}{a_f n + \gamma c w^2(\bar{a} + \bar{b}n)}}{-\frac{\gamma n}{2} + \frac{\gamma c w^2 n}{a_f n + \gamma c w^2(\bar{a} + \bar{b}n)}} \right)\end{aligned}$$

Now, if we write $r = a_f + \gamma c \omega^2 \bar{b}$ and $q = \gamma c \omega^2 \bar{a}$, then,

$$\beta = \begin{pmatrix} \frac{-\gamma}{2} + \frac{\gamma c w^2}{r n + q} \\ \frac{-\gamma n}{2} + \frac{\gamma c w^2 n}{r n + q} \end{pmatrix}$$

From here, we want to calculate $E(\beta)$, assuming that nutriation n follow an expo-

nential distribution. Then,

$$\begin{aligned}
\mathbb{E} \left(\frac{\partial \ln \bar{W}(n, \bar{y})}{\partial \bar{a}} \right) &= \frac{-\gamma}{2} + \gamma c w^2 \mathbb{E} \left(\frac{1}{r n + q} \right) \\
&= \frac{-\gamma}{2} + \gamma c w^2 \int_0^\infty \frac{1}{r n + q} \lambda \exp(-\lambda n) dn \\
&= \frac{-\gamma}{2} + \frac{\lambda \gamma c w^2}{r} \exp \left(\frac{\lambda q}{r} \right) E_1 \left(\frac{\lambda q}{r} \right) \\
\mathbb{E} \left(\frac{\partial \ln \bar{W}(n, \bar{y})}{\partial \bar{b}} \right) &= \frac{-\gamma \lambda}{2} + \gamma c w^2 \mathbb{E} \left(\frac{n}{r n + q} \right) \\
&= \frac{-\gamma \lambda}{2} + \gamma c w^2 \int_0^\infty \left(\frac{n}{r n + q} \right) \lambda \exp(-\lambda n) dn \\
&= \frac{-\gamma \lambda}{2} + \gamma c w^2 \left(\frac{1}{r} - \frac{\lambda q}{r^2} \exp \left(\frac{\lambda q}{r} \right) E_1 \left(\frac{\lambda q}{r} \right) \right),
\end{aligned}$$

where $E_1(x) = \int_x^\infty \frac{\exp(-t)}{t} dt$. From here, we can calculate $\bar{a}$ and $\bar{b}$ numerically by setting $E(\beta) = 0$. In other words,

$$\begin{aligned}
\frac{-\gamma}{2} + \frac{\lambda \gamma c w^2}{r} \exp \left(\frac{\lambda q}{r} \right) E_1 \left(\frac{\lambda q}{r} \right) &= 0 \\
\frac{-\gamma \lambda}{2} + \gamma c w^2 \left(\frac{1}{r} - \frac{\lambda q}{r^2} \exp \left(\frac{\lambda q}{r} \right) E_1 \left(\frac{\lambda q}{r} \right) \right) &= 0
\end{aligned}$$

From the first expression, we get:

$$\exp \left(\frac{\lambda q}{r} \right) E_1 \left(\frac{\lambda q}{r} \right) = \frac{r}{2 \lambda c w^2}$$

Substituting this to the second equation, we get:

$$\begin{aligned}
\frac{-\gamma \lambda}{2} + \gamma c w^2 \left(\frac{1}{r} - \frac{q}{2 r c w^2} \right) &= 0 \\
\frac{-\gamma \lambda}{2} + \gamma \left(\frac{c w^2}{r} - \frac{q}{2 r} \right) &= 0
\end{aligned}$$

Equivalently, by cancelling for γ and multiplying by $2r$, we get:

$$\lambda r = 2 c w^2 - q.$$

Now, if we write $x = \lambda q / r$, then

$$x = \frac{\lambda q}{r} = \frac{2 c w^2}{r} - \lambda^2$$

which suggests:

$$r = \frac{2 c w^2}{x + \lambda^2}.$$

Thus,

$$\exp(x) E_1(x) = \frac{1}{x + \lambda^2},$$

which can be numerically solved for x . Provided that x is known, we have:

$$r = \frac{2c\omega^2}{x + \lambda^2}, q = \frac{2c\omega^2 x}{\lambda^2 + x}$$

Finally, this gives us $\bar{a}$ and $\bar{b}$:

$$\begin{aligned}\bar{a} &= \frac{2x}{\gamma(\lambda^2 + x)} \\ \bar{b} &= \frac{r - a_f}{\gamma c \omega^2} = \frac{2\lambda}{\gamma(\lambda^2 + x)} - \frac{a_f}{\gamma c \omega^2}\end{aligned}$$